King Saud University
College of Computer and Information Sciences
Department of Information Technology
IT 362: Principles of Data Science
Project – Phase#1

Saudi Law project

LOGBOOK

Phase#1

First Semester 2025

Project Overview

Saudi Labor Law Q&A Dataset
This project aims to build a structured dataset of the Saudi Labor Law, including its official articles and frequently asked questions (FAQs), in order to make the information more accessible for analysis, visualization, and potential applications such as question - answering systems.

The dataset captures:
• Articles of the law (raw legal text).
• FAQs (question - answer pairs).
• Metadata linking each entry to its source URL or official PDF.

This ensures that the dataset can be used in future machine learning and data science projects that require legal text processing in Arabic.

Data Collecting

Data Collection Period: [10 September 2025] To [24 September 2025]

Data sources :

- Bureau of Experts at the Council of Ministers(As pdf) : The official text of the Saudi Labor Law.

Link :http://laws.boe.gov.sa/BoeLaws/Laws/LawDetails/08381293-6388-48e2-8ad2-a9a700f2aa94/1

- Istitlaa Platform (By web scrapping) : Draft amendments to the Saudi Labor Law regulations by MHRSD on the Istitlaa platform.

Link:https://istitlaa.ncc.gov.sa/ar/Labor/Hrsd/Regulations5/Pages/default.aspx

- Qiwa Platform (By web scrapping) : Qiwa is an MHRSD digital platform offering labor law content and electronic services for private-sector employers and workers.

Link :https://qiwa.sa/ar/labor-law

- Master’s Theses from King Saud University Digital Library(As pdf): Master’s theses containing analytical studies on Saudi labor law.

Link:https://drive.google.com/drive/folders/1wLQwAG_AKiWEzeCQFQtK6wy6IhCuZFrG

These sources were selected because they represent official and reliable references that ensure the accuracy and validity of the legal data, which enhances the credibility of the project and the quality of its outcomes. In addition, these sources supported by explanatory materials and simplified FAQs that help bridge the gap between complex legal wording and practical understanding. This combination of precise legal texts and clarifying explanations enables us to obtain a comprehensive and balanced dataset, making it easier to process and analyze in future applications such as question–answer systems.

Challenges and Actions taken:

Website Structure :

- Qiwa presents the Saudi Labor Law in separate sections, each under a unique URL. This made scraping more complex because the data was not centralized on a single page.
- Solution: A CSV file was created with: Column 1 = URL of the section Column 2 = Extracted text content This ensured the data was organized and traceable to its source.

Different Data Formats :

- The law text and the FAQ section had different structures. The law text was straightforward, while FAQs required a question–answer format.
- Solution: FAQ data was stored in a separate CSV file (labor_law_faq.csv) with: Column 1 = Question Column 2 = Answer This separation ensured consistency and ease of later processing.

WebDriver Configuration :

- Setting up Selenium WebDriver for headless scraping posed a challenge.
- Solution: Proper configuration was required to ensure compatibility with dynamic content loading and to allow scraping without opening a browser window. This involved fine-tuning browser options for performance and stability.

Prevent web scraping :

- Some websites prevent or hinder automated data collection by using dynamic JavaScript loading, protections like CAPTCHA, restrictions in the robots.txt file, or automated behavior detection systems (rate-limiting, WAF). This caused data collection interruptions, incomplete results, repeated errors during execution, and even the risk of account suspension or IP blocking.
- Solution: Shifted to sources that allow web scraping and ensured proper permissions and legal compliance.

Text Encoding :

- Arabic text initially appeared corrupted in Excel.
- Solution: Exported the files with UTF-8-SIG encoding to ensure proper display.

Web Scraping

Web Scraping Tools Used

- Selenium : Interacting with dynamic web pages and handling JavaScript-based content.
- webdriver_manager : Automatically installs and manages ChromeDriver.
- pandas : Organizing and saving the scraped data in tabular format.
- csv : Writing extracted results into CSV files.
- Requests (common tool, not used in our script) : Sending HTTP requests to fetch static web content.
- BeautifulSoup (common tool, not used in our script) : Parsing and extracting information from HTML documents.

Libraries, Modules, and Methods Used in the Web Scraping

Libraries Used

- selenium : Web scraping and interacting with dynamic web pages.
- webdriver_manager : Automatically installs and manages ChromeDriver.
- pandas : Data processing and saving results into CSV.
- csv : Writing data directly into CSV files.
- os : File and path management.
- time : Adding delays (e.g., sleep) during page loading.

Selenium Modules Used

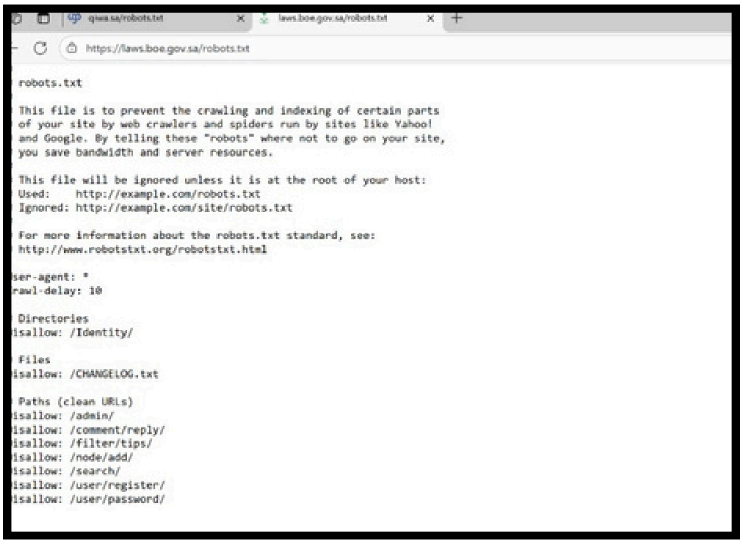
- webdriver : Launch and control the browser.
- By : Locate elements using CSS selector, XPath, ID, etc.
- WebDriverWait : Wait until a specific condition is met.(the page is fully loaded)
- expected_conditions (EC) : Predefined conditions used with WebDriverWait (e.g., presence, visibility).

Methods Used

- driver.get(url) : Open the target web page.
- driver.find_element(...) / driver.find_elements(...) :Find element(s) on the page.
- element.click() : Click on an element (e.g., expand accordion for answers).
- element.text : Extract text content from an element.
- wait.until(EC....) :Wait until a condition is met (e.g., element is present).
- EC.presence_of_element_located(locator) : Check that the element exists in the DOM.
- EC.visibility_of_element_located(locator) : Check that the element is visible before interacting.
- list.append(item) : Store extracted results in a list.
- csv.writer.writerow(row) : Write one row into a CSV file.
- pandas.DataFrame.to_csv(path, ...) : Save results into a CSV file.

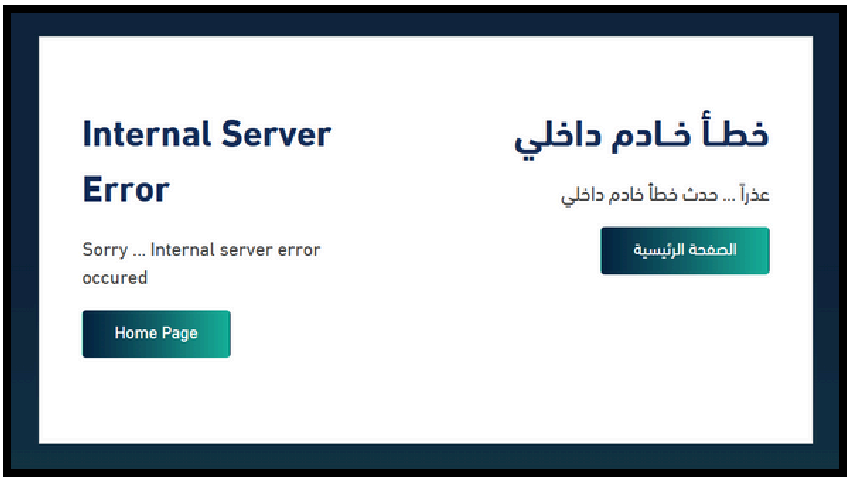
Robots.txt:

1- Bureau of Experts at the Council of Ministers




```
robots.txt

This file is to prevent the crawling and indexing of certain parts
of your site by web crawlers and spiders run by sites like Yahoo!
and Google. By telling these "robots" where not to go on your site,
you save bandwidth and server resources.

This file will be ignored unless it is at the root of your host:
Used:    http://example.com/robots.txt
Ignored: http://example.com/site/robots.txt

For more information about the robots.txt standard, see:
http://www.robotstxt.org/robotstxt.html

ser-agent: *
rawl-delay: 10

Directories
isallow: /Identity/

Files
isallow: /CHANGELOG.txt

Paths (clean URLs)
isallow: /admin/
isallow: /comment/reply/
isallow: /filter/tips/
isallow: /node/add/
isallow: /search/
isallow: /user/register/
isallow: /user/password/
```

2- Istitlaa Platform



The website didn't allow to enter the page.

3- Qiwa Platform

Saudi Law project

LOGBOOK

Phase#2

Data cleaning and preprocessing:

Overview:

The cleaning and preprocessing stage was essential to prepare the four collected datasets: Qiwa Articles, Qiwa FAQ, BOE Official Law, and Istitlaa (Public Consultation), for further analysis.
This phase ensured the removal of noise, correction of formatting inconsistencies, and standardization of text across all datasets.
Both cleaning and preprocessing were implemented using Python, primarily with the pandas and re (regular expressions) libraries.

`1.BOE (Official Law) Dataset:

Steps Performed

1. Imported the dataset (boe_laborlaw_articles_Edit.csv) using UTF-8-SIG encoding.
2. Created a copy to preserve the raw data before modifications.
3. Executed the following cleaning operations:
• Removed duplicate entries using drop_duplicates().
• Filled missing cells with the string "N/A" for consistency.
• Applied a custom function clean_text() to every text-based column to:
• Convert text to lowercase.
• Remove special characters such as colons (:).
• Replace multiple line breaks, tabs, or spaces with a single space.
4. Saved the cleaned file as boe_cleaned.csv.

Purpose

• Legal text often contains formatting inconsistencies, line breaks, and punctuation marks that interfere with automated text analysis.
• Lowercasing the text and removing unnecessary characters helped create a unified format for subsequent natural language processing steps.
• Filling missing cells prevented processing errors in future stages (e.g., tokenization, length analysis).

Preprocessing Aspect

• Text normalization: converting all text to a uniform lowercase structure.
• Whitespace standardization: ensuring each entry maintains consistent spacing.
• Encoding consistency: Arabic text preserved without distortion.

Libraries Used

• pandas for data manipulation.
• re for regex-based cleaning and whitespace normalization.

2. Istitlaa (Public Consultation) Dataset

Steps Performed

1. Loaded the dataset (istitlaa_laborlaw.csv) using UTF-8-SIG encoding.
2. Duplicates were removed using drop_duplicates().
3. Missing values were replaced with "N/A".
4. Applied the same clean_text() function used in BOE cleaning to standardize and normalize all text fields.
5. Exported the cleaned data as istitlaa_cleaned.csv.

Purpose

• The dataset contained irregular formatting due to data extraction from online consultation pages.
• Cleaning removed extra spaces, invisible characters, and inconsistent punctuation.
• Replacing missing values ensured structural integrity across the file, allowing comparisons with other datasets.

Preprocessing Aspect

• Normalized Arabic text.
• Reduced unnecessary noise for consistent analysis.
• Ensured comparability with other datasets during the EDA phase.

Libraries Used

• pandas
• re
• UTF-8-SIG encoding support.

3. Qiwa Articles Dataset

Steps Performed

1. Loaded the dataset (qiwa_data.csv) which contained a single column merging both URLs and content due to the web-scraping process.
2. Implemented a custom function (clean_qiwa_data) that:

- Read the file line by line to correctly separate URLs from their associated content.
- Grouped all fragmented text belonging to the same URL into a single cell.
- Identified the "canceled" chapter (Chapter 13) and replaced all related articles with a standardized cancellation note.
- Applied a duplication check using a custom logic that detected and removed major repeated text blocks.

3. Saved the cleaned version as qiwa_data_cleaned.csv encoded in UTF-8-SIG for Arabic compatibility.

Purpose

- The scraping process had merged links and content together, so separating them ensured each record represented one unique article.
- Grouping text by URL prevented fragmented information.
- Removing repeated or canceled articles eliminated redundant and invalid data, ensuring the dataset represented valid legal articles only.
- Applying UTF-8-SIG encoding prevented corruption of Arabic letters.

Preprocessing Aspect

- During cleaning, the function implicitly performed preprocessing:
- Normalized text structure (removed duplicated lines, unified formatting).
- Created a consistent representation of legal articles ready for textual analysis.

Libraries Used

- pandas for reading, grouping, and exporting CSVs.
- re for pattern matching in duplication detection.
- Standard Python file operations (open, readlines) for structured data reconstruction.

4. Qiwa FAQ Dataset

Steps Performed

1. Loaded the dataset (labor_law_faq.csv) containing two columns: Question and Answer.
2. Checked for structure consistency, ensuring both columns existed and contained valid string entries.
3. Applied a cleaning function to:

- Remove all line breaks (\n, \r).
- Replace multiple whitespaces with a single space.
- Strip unnecessary spaces from the beginning and end of text.

4. Exported the cleaned data as labor_law_faq_cleaned.csv using UTF-8-SIG encoding.

Purpose

- The presence of newline characters and uneven spacing made text processing unreliable (e.g., incorrect token counts).
- By unifying spacing and removing line breaks, questions and answers could be processed as clean, single-line records.
- Ensuring consistent format across all entries was critical before running frequency or topic analyses.

Preprocessing Aspect

- Text normalization (standardizing spacing, punctuation, and encoding).
- Ensuring proper token separation for later NLP analysis.

Libraries Used

- pandas for data manipulation.
- re for regex-based whitespace cleanup.

Data cleaning and preprocessing outcomes:

After cleaning and preprocessing, all datasets were unified in structure, consistent in encoding, and free from duplicates or formatting noise.
This provided a solid, reliable foundation for the next stage: Exploratory Data Analysis (EDA) and Comparison.

Exploratory Data Analysis (EDA):

Overview:
After cleaning and preprocessing the four datasets (Qiwa Articles, Qiwa FAQ, BOE Official Law, and Istitlaa Public Consultation), the next step was to perform Exploratory Data Analysis (EDA).
This phase aimed to uncover key patterns, differences, and trends in text length, structure, and content themes across datasets.
EDA was implemented using Python, primarily through pandas, matplotlib, and seaborn, along with basic statistical functions and text metrics.

1. Loading and Preparing Data

- The cleaned datasets (qiwa_data_cleaned.csv, labor_law_faq_cleaned.csv, boe_cleaned.csv, and istitlaa_cleaned.csv) were imported using pandas.
- Each dataset was assigned a clear variable name to maintain traceability (e.g., qiwa_articles, faq, boe, istitlaa).
- The text columns were checked for proper encoding and inspected to ensure no further missing or null values remained.

Purpose:
This step ensured that the data used for analysis was consistent, error-free, and properly formatted after the cleaning phase.

2. Descriptive Statistics and Structure Analysis

Steps Performed

1. Calculated key text metrics for each dataset:
   - Total number of rows (entries).
   - Average text length (mean) based on character count.
   - Median text length (median) to represent central tendency.
2. Stored these metrics in a pandas DataFrame for structured comparison.
3. Used the DataFrame to generate tables that display differences in text size and structure.

Purpose

- Calculating mean and median text lengths helps understand whether each dataset contains concise or lengthy content.
- The number of rows reveals the scope and coverage of each data source (e.g., how extensive or brief it is).
- This provides a quantitative foundation for comparing datasets later.

Example Observations

- Qiwa (FAQ) contained 32 entries with very short texts.
- Qiwa (Articles) showed extremely long records.
- BOE included a moderate amount of text, while Istitlaa showed mid-length entries.

Libraries Used

- pandas for statistical computations and summaries.
- numpy for numerical calculations where applicable.

3. Topic and Keyword Exploration

Steps Performed

1. Extracted the most frequent words/topics within each dataset using simple token-based counting.
2. Displayed results in tables showing the top repeated words for each source.
3. Visualized term frequency distributions to identify common themes such as “work”, “law”, “article”, and “rights”.

Purpose

- Identifying frequently used words helps capture the dominant topics in each dataset.
- It also supports domain understanding showing how legal, procedural, or user-centered each dataset is.
- Frequent keywords highlight thematic overlap between datasets (e.g., all datasets centered on labor and regulation).

Interpretation

- Qiwa FAQ focused on direct practical terms like “employee”, “rights”, and “contract”.
- Qiwa Articles and BOE emphasized legal terminology such as “article”, “law”, and “decision”.
- Istitlaa featured consultation-related words such as “proposal” and “training”, showing its participatory nature.

Libraries Used

- pandas for grouping and counting tokens.
- collections.Counter for frequency computation (if used).
- matplotlib / seaborn for visualizing top keywords.

4. Visualization of Text Distributions

Steps Performed

1. Created histograms and boxplots to visualize the distribution of text lengths for each dataset.
2. Compared how skewed or balanced each dataset was (e.g., some had many short entries, others had long ones).
3. Created a pie chart for visual comparison of the median text lengths across all four datasets.

Purpose

- Visualization helps quickly identify which dataset contains longer or shorter content.
- Pie and box plots allow intuitive comparison of distribution patterns.
- Confirms quantitatively observed differences through graphical representation.

Interpretation

- Qiwa Articles dominated with the highest median and mean length.
- FAQ entries were shortest, consistent with their Q&A purpose.
- BOE and Istitlaa positioned in between, offering structured but not overly lengthy text.
- This supports the idea that dataset purpose directly influences its text complexity and size.

Libraries Used

- matplotlib.pyplot for line, bar, and pie charts.
- seaborn for aesthetic histograms and boxplots.
- pandas built-in plotting functions for quick visual summaries.

Comparison of Datasets (Cross-Analysis)

After completing the EDA for each dataset, a comparison was conducted to summarize key differences in structure, text length, and topic focus.
The detailed numerical results and interpretations are documented in the main project report.

Steps Performed

1. Combined summary metrics from all datasets (mean, median, number of rows, top topics) into one comparative table.
2. Sorted and formatted the table to clearly highlight differences.
3. Plotted the median comparison using a pie chart for visual clarity.
4. Wrote detailed interpretations explaining the observed hierarchy of datasets.

Purpose

- Comparing datasets without merging allows discovering contextual differences.
- Helps identify which dataset is more detailed, concise, or targeted to specific audiences.
- Supports later validation of results and interpretation of domain patterns.

Interpretation Summary

- Qiwa (FAQ) → concise, user-centered answers.
- Qiwa (Articles) → long, detailed, official documents.
- BOE → structured legislative text, moderate in length.
- Istitlaa → medium-sized proposals focused on consultation.
- Shared themes include “work”, “article”, and “law”, but with differing depth and focus.

Libraries Used

- pandas for tabular comparison.
- matplotlib for pie chart and visual summaries.

Challenges and Actions taken:

Challenge	Description	Action Taken
Handling different text structures in Qiwa dataset	During the EDA phase, one teammate faced difficulty analyzing Qiwa data because it contained two distinct formats: FAQ (Question–Answer pairs) and Articles (long-form legal text). Combining them distorted the results since the structure and purpose differed.	The datasets were separated and analyzed independently. Each subset (FAQ and Articles) underwent its own descriptive and visual analysis, ensuring accuracy and clearer insights.
Incorrect HTML parsing for BOE dataset	When inspecting the raw scraped text, one member noticed missing or misaligned columns due to incorrect HTML parsing. Headings such as “Chapter One” and “Article One” were not properly identified, causing shifted data.	The HTML structure was re-read and redefined, explicitly marking each section’s start using text patterns like “Chapter One” and “Article One.” This restored the proper column alignment before further processing.
Unbalanced dataset sizes	The Qiwa Articles dataset was much larger than the others, which caused scaling issues in graphs and skewed averages.	Used median values instead of means for fairer comparison, and normalized visualization axes to handle range differences.
Master PDF files were unstructured and unsuitable for automated text extraction	During the data cleaning and exploratory analysis phase, the team identified that several master documents were provided only in PDF format with complex academic layouts, mixed Arabic–English text, and inconsistent formatting. This made it impossible to convert them into structured CSV or analyzable form using Python libraries.	The team excluded the master PDFs from the EDA process and focused on web-based and CSV data sources to ensure clean, consistent, and analyzable data.

Saudi Law project

LOGBOOK

Phase#3

Plan the modelling and evaluation strategy for the Legal Q/A system.

Tasks:

1. Identify modelling task type
    - Our project is Legal Question–Answering over Saudi Labor Law.
    - The task is not a classic classification/regression, and not pure clustering.
    - Instead, it is a combination of:
        - Information Retrieval (baseline model using BM25).
        - Generative Q/A Models (LLM prompting with different few-shot setups).
2. Define models to be evaluated in Phase 3

We decided to evaluate four configurations:

Name	Type	Description
Baseline Retrieval (BM25)	Non-generative IR baseline	Returns top law articles based on keyword matching.
Zero-Shot	Generative LLM	Gemini Pro answers without examples.
One-Shot	Generative LLM	Gemini Pro with 1 example Q/A.
Three-Shot	Generative LLM	Gemini Pro with 3 example Q/A (few-shot prompting).

3.Define evaluation dimensions

- Objective metrics (token overlap, coverage, relevance).
- Subjective (manual) metrics (correctness, grammar, style).
- Specific focus on numerical/legal rules (e.g., wage deductions, disciplinary articles).

Decisions & Rationale

- Instead of using classification/regression algorithms, we treat the problem as:
  - Baseline retrieval model vs LLM-based generative models.
- We commit to:
  - 1 baseline model (BM25),
  - 3 generative prompting setups (Zero / One / Three-Shot),
  - Detailed objective + manual evaluation.

Building the LLM-Generated Q/A Dataset

Goal: Prepare a unified Q/A dataset used for evaluation of the baseline model and to support Phase 3 experiments.

Input Datasets

We used the following cleaned datasets from Phase 2:
File Name - Description:
boe_cleaned.csv-Cleaned articles from BOE (official law text).
istitlaa_cleaned.csv-Drafts / consultations related to labor law.
qiwa_data_cleaned.csv-Qiwa platform data (structured legal info).
labor_law_faq_cleaned.csv-Frequently asked questions about labor law.
Role of each dataset:

- BOE → Main legal reference (ground truth articles).
- Istitlaa → Additional legal context.
- Qiwa + FAQ → Natural-language explanations and practical Q/A.

Generating Q/A Pairs with Gemini 2.5 Pro

We used Gemini 2.5 Pro to generate a combined labor_law_qa.csv that contains Q/A pairs grounded in the above datasets.

- For each source dataset, Gemini is prompted to:
  - Read the given article or legal text.
  - Generate questions that a worker or employer might ask.
  - Generate legally correct answers based on the same text.

- Final dataset:
  - Total samples: 321 Q/A pairs.
  - Sources mixed: boe_cleaned.csv, istitlaa_cleaned.csv, qiwa_data_cleaned.csv, labor_law_faq_cleaned.csv.
  - Columns include: Question, Answer, Source, Topic, etc.

Summary of All Models (Baseline, Zero-Shot, One-Shot, Three-Shot)

1. Baseline Model (BM25 Retrieval)

The baseline model uses BM25, a keyword-based retrieval algorithm that returns the most relevant legal articles purely based on text similarity.
It does not generate answers; instead, it retrieves the closest matching article.

What happened:

- Built a unified corpus from BOE, Qiwa, Istitlaa, and FAQ datasets.
- Tokenized all texts and indexed them using BM25.
- Evaluated on 65 test questions (stratified sample).
- Performance was very low in Jaccard and Coverage because the baseline returns raw law text, while ground-truth answers are natural-language explanations.

Conclusion:
A useful lower-bound benchmark, but insufficient for legal Q/A tasks.

2. Zero-Shot Model

Gemini answered each question without any examples.

What happened:

- Generated 267 Q/A pairs.
- Moderate lexical similarity to BOE article text (Jaccard ≈ 0.14).
- Coverage ≈ 0.32, meaning about one-third of answer tokens appeared in the real article.
- Often missing article numbers.
- Struggled with numeric rules (e.g., suspensions, deductions, deadlines).
- Language sometimes vague and uncertain (“may”, “often”).

Conclusion:
Unstable and not reliable for precise legal content.

3. One-Shot Model

Gemini received one example Q/A before generating answers.

What happened:

- Generated 207 Q/A pairs.
- Similar overlap to Zero-Shot but more structured and clearer.
- Almost always cited article numbers.
- Best Arabic grammar and clarity.
- Sometimes paraphrased too much or added unnecessary commentary.
- Still struggled with certain strict numeric rules.

Conclusion:
Much better than Zero-Shot; linguistically strong, but not the most legally precise.

4. Three-Shot Model

Gemini received three example Q/A pairs (few-shot prompting).

What happened:

- Generated 426 Q/A pairs.
- Highest coverage (≈ 0.39), meaning answers contain more legally relevant content.
- Most consistent mapping between questions and correct BOE articles.
- Best accuracy on sensitive articles:
  - Disciplinary rules (78–83),
  - Wage deductions (98–100),
  - Applicability of law (5),
  - Wage payment (97).
- Lowest hallucination and best overall stability.

Conclusion:

The best-performing model across correctness, grounding, and stability.

Evaluation Summary

We evaluated four configurations to determine the best model for Saudi Labor Law Q/A: Baseline, Zero-Shot, One-Shot, and Three-Shot.

The evaluation used objective metrics (Jaccard, Coverage, Relevance) and manual metrics (Correctness, Grammar, Style).

Objective Metrics Used

- Jaccard Overlap: measures word-level similarity with the correct article.
- Coverage (Answer in Article): how much of the model answer appears in the real BOE article text.
- Coverage (Article in Answer): how much of the law appears in the generated answer.
- Relevance Score: percentage of answers meeting minimum similarity thresholds.
- Answer Length: detects missing details or hallucinations.

Manual Metrics Used

- Correctness (0–2): legally accurate or not.
- Grammar (1–5): quality of Arabic writing.
- Style (0–2): legal clarity and consistency.

Overall Findings

- Baseline: Lowest scores; retrieves raw text only.
- Zero-Shot: Unstable, vague, struggles with numeric rules.
- One-Shot: Best grammar, clearer structure, but sometimes paraphrased too much.
- Three-Shot: Best legal correctness, highest grounding, most stable answers.

Challenges & Issues

Initial Approach Was Too Complex:
We originally planned to build the LLM system using a full RAG pipeline (retrieval, embeddings, vector database, chunking, hybrid search).
But this approach was:

- Technically complex
- Time-consuming
- Difficult to tune
- Too heavy for the project timeline

So we changed direction and focused on prompt-based generation (Zero/One/Three-Shot) which produced better and faster results.

Article Matching Issues:
Some generated Q/A pairs did not match the correct BOE article due to:

- Differences in formatting
- Variations in writing “المادة ...”
- Missing or ambiguous article numbers
- We solved this by creating a normalization function for article labels.

Sensitive Articles Required Extra Attention:
Articles with numbers or deadlines (e.g., 78–83, 98–100) caused:

- Frequent mistakes in Zero-Shot
- Partial correctness in One-Shot
- They required manual review and special rules to validate.

Time Constraints:
Evaluating hundreds of Q/A pairs required:

- Careful merging
- Tokenization fixes
- Manual checking

Balancing accuracy with time was challenging.